

Lecture 08: Types of Malware

Malware

Malware (Malicious Software) is any software intentionally designed to damage, disrupt, steal data from, or gain unauthorized access to a computer system or network. The main objective of malware is to compromise the Confidentiality, Integrity, and Availability (CIA) of information systems.

Characteristics of Malware

- Developed for malicious purposes.
- Runs without the user's permission.
- May steal, destroy, or modify data.
- Can spread automatically or require user interaction.
- Often attempts to hide from detection.

Classification of Malware

Malware is broadly classified into two categories.

1. Malware that Needs a Host Program

These malware types attach themselves to another legitimate program and execute when that program is run.

Examples:

- Trap Door (Backdoor)
- Logic Bomb
- Trojan Horse
- Browser Plug-ins/Extensions/Scripts
- Virus

2. Independent Malware

These malware types do not require another host program and can execute independently.

Examples:

- Worm
- Advanced Persistent Threat (APT)
- Botnet

Trap Door (Backdoor)

A Trap Door (also called a Backdoor) is a secret entry point into a computer program or system that allows someone to bypass normal authentication and security mechanisms. Developers may intentionally create backdoors for testing or maintenance, but attackers often exploit or install them to gain unauthorized access.

Working Principle

1. System contains a hidden access method.
2. Attacker knows the special password or input sequence.
3. Normal authentication is bypassed.
4. Attacker gains unauthorized access.

Characteristics

- Hidden entry point.
- Bypasses authentication.
- Difficult for users to detect.
- Provides administrator-level access in many cases.

Advantages (for Attackers)

- Easy system access.
- Avoids normal login.
- Long-term control.

Prevention

- Remove unnecessary backdoors.
- Update software regularly.
- Conduct security audits.
- Use intrusion detection systems.

Logic Bomb

A Logic Bomb is malicious code embedded inside a legitimate program that remains inactive until a specific condition or event occurs. When the condition is satisfied, it executes harmful actions.

Trigger Conditions

Examples include:

- A specific date or time.
- User login.
- File deletion.
- Program execution.
- Employee termination.

Working Process

Malicious Code Hidden

↓

Condition Not Met

↓

Remains Dormant

↓

Condition Satisfied

↓

Executes Attack

Examples

- Delete all files on a certain date.
- Format a hard drive after 100 logins.
- Shut down a server after an employee leaves.

Prevention

- Code review.
- Software auditing.

- File integrity monitoring.
- Antivirus scanning.

Trojan Horse

A Trojan Horse is malware made to look like a useful or legitimate application that performs hidden malicious actions when executed. Unlike viruses and worms, Trojans do not replicate themselves.

Working Process

1. User downloads a seemingly useful application.
2. User executes the program.
3. Hidden malicious code runs.
4. Sensitive information is stolen or the system is compromised.

Examples

- Fake antivirus software.
- Cracked games.
- Fake software updates.
- Fake banking applications.

Characteristics

- Appears legitimate.
- Requires user execution.
- Does not self-replicate.
- Often installs additional malware.

Prevention

- Download software from trusted sources.
- Verify digital signatures.
- Keep antivirus updated.
- Avoid pirated software.

Virus

A Computer Virus is malware that attaches itself to another executable program and spreads by modifying that program. Viruses require human action, such as opening an infected file, to spread.

Characteristics

- Needs a host program.
- Self-replicates.
- Modifies existing programs.
- Can corrupt or destroy files.

Virus Life Cycle

A virus generally passes through four phases.

1. Dormant Phase

The virus remains inactive and waits for a trigger.

2. Propagation Phase

The virus copies itself into other programs or files.

3. Triggering Phase

A specific condition activates the virus.

Examples:

- Opening an email attachment.
- Specific date.
- Running a program.

4. Execution Phase

The virus performs its malicious activity.

Examples:

- Delete files.
- Corrupt data.
- Display messages.
- Slow down the computer.

Virus Structure

A typical virus program contains:

1. Entry Point

Transfers control to the virus.

2. Infection Flag

Checks whether a program is already infected.

3. Infection Routine

Searches for uninfected files.

4. Payload

Performs malicious actions.

5. Return Control

Transfers execution back to the original host program to avoid suspicion.

Some viruses also compress or decompress host programs and monitor program size to avoid detection.

Types of Viruses

1. Parasitic Virus

Attaches itself to executable programs and infects other executable files.

2. Memory-Resident Virus

Loads into RAM and infects programs while they are running.

3. Macro Virus

Embedded inside office documents.

Runs automatically when documents are opened.

4. Boot Sector Virus

Infects the boot sector.

Runs every time the computer starts.

5. Polymorphic Virus

Changes its internal code or encrypts itself with a different key each time it spreads.

Purpose: Avoid antivirus signature detection.

Boot Sector Virus

A Boot Sector Virus infects the boot sector or Master Boot Record (MBR) of a storage device. It executes immediately when the computer starts.

Characteristics

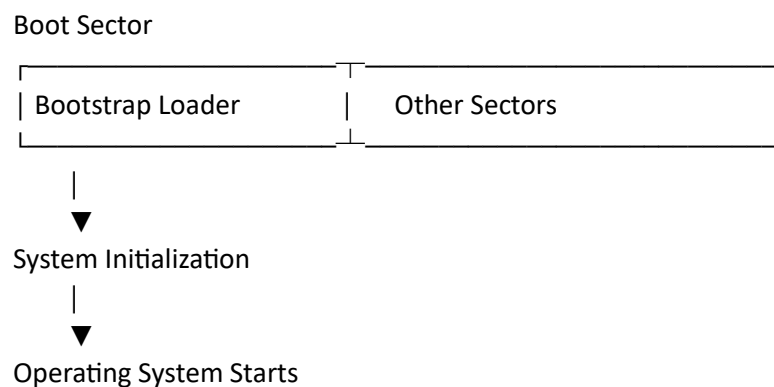
- Executes before the operating system loads.
- Difficult to remove.
- Can prevent system startup.
- Infects removable storage devices.

Prevention

- Disable booting from unknown USB drives.
- Use updated antivirus software.
- Protect the boot sector.

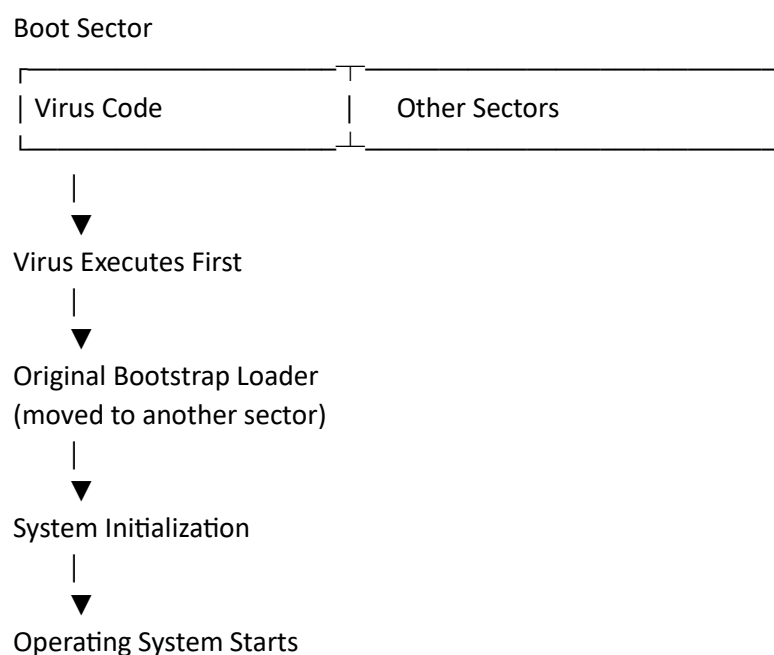
(a) Before Infection

Diagram



(b) After Infection

Diagram



Macro Virus

A Macro Virus is malware written using the macro programming language of applications such as Microsoft Word or Microsoft Excel. It spreads through infected documents.

How Macro Virus Spreads

1. Virus macro is attached to a Word document.
2. User opens the document.
3. Macro executes automatically.
4. Virus copies itself to the global macro template.
5. Every newly opened document becomes infected.

Advantages (for Attackers)

- Easy distribution through email.
- Difficult for ordinary users to notice.
- Cross-platform document sharing.

Prevention

- Disable automatic macros.
- Open trusted documents only.
- Keep Office software updated.

Rootkit

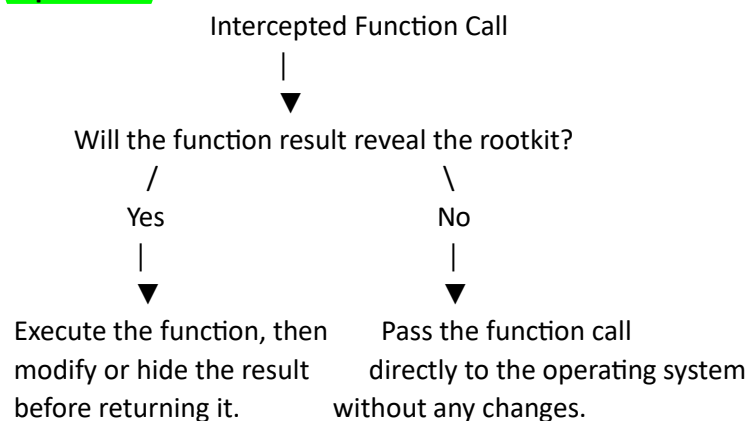
A Rootkit is malware that hides itself and other malware from the operating system, users, and security software. It modifies operating system components to remain invisible.

Characteristics

- Resides inside the operating system.
- Modifies OS code.
- Hides malicious files.
- Hides malicious processes.
- Avoids antivirus detection.

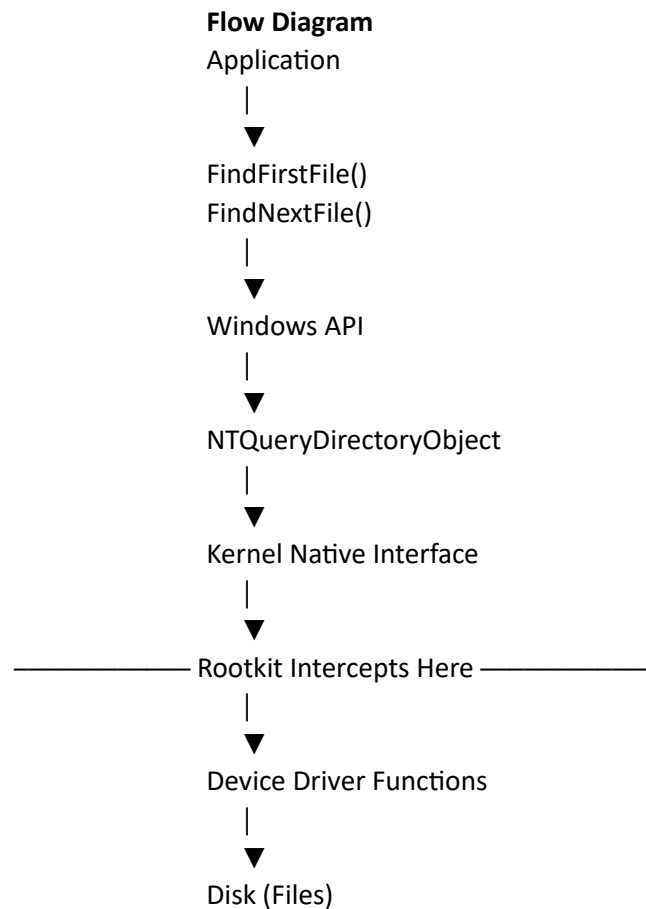
This flowchart explains how a rootkit hides itself by intercepting operating system function calls.

Explanation



This diagram shows how a rootkit hides files by intercepting Windows file access requests.

Rootkit File Hiding Process



Functions

Rootkits can hide:

- Files.
- Processes.
- Registry entries.
- Network connections.

Example:

Hidden from commands such as

ls

or

ps

Prevention

- Secure Boot.
- System integrity monitoring.
- Boot-time antivirus scanning.
- Regular operating system updates.

Worm

A Computer Worm is malware that spreads automatically through computer networks without requiring a host program or user interaction. Unlike viruses, worms are independent.

Working Process

1. Finds vulnerable systems.
2. Infects them automatically.
3. Copies itself.
4. Repeats continuously.

Characteristics

- Self-replicating.
- Uses network connections.
- Consumes network bandwidth.
- No user action required.

Virus vs Worm

Virus	Worm
A virus is malicious software that attaches itself to a host program or file.	A worm is a standalone malicious program that does not need a host program.
Requires a host program or file to run.	Runs independently without a host program.
Requires user action (e.g., opening an infected file or program) to spread.	Spreads automatically without user action.
Mainly spreads through infected files, email attachments, USB drives, or downloaded software.	Mainly spreads through computer networks by exploiting vulnerabilities.
Infects and modifies files or programs.	Infects computers and networks but usually does not modify files.
Spreads relatively slowly because it depends on user actions.	Spreads very quickly because it replicates automatically over networks.
Usually damages or corrupts files, programs, or system data.	Mainly consumes network bandwidth and system resources, causing slowdowns or network congestion.
Examples: CIH (Chernobyl), Melissa, ILOVEYOU.	Examples: Conficker, SQL Slammer, WannaCry (worm component).

Morris Internet Worm

The Morris Worm (1988) was the first Internet worm that spread rapidly across the Internet. It unintentionally caused massive disruption because of a programming error that allowed repeated infections of the same machines.

Objectives

- Find vulnerable computers.
- Spread automatically.
- Remain hidden.

Effects

- Resource exhaustion.
- Network congestion.
- Server crashes.

- Internet disruption.

Techniques Used

The Morris Worm exploited:

- Weak passwords.
- Buffer overflow in fingerd.
- Backdoor in sendmail.

It also loaded its code into memory, encrypted parts of itself, changed process names and IDs periodically, and removed files to remain difficult to detect.

Lessons Learned

- Regular security patching.
- Vulnerability scanning.
- Creation of the Computer Emergency Response Team (CERT).

Malware Detection and Prevention

Prevention

Best prevention practices include:

- Limit unnecessary Internet exposure.
- Keep software updated.
- Install antivirus software.
- Use firewalls.
- Download software only from trusted sources.
- Train users against phishing attacks.

Malware Detection Approaches

The lecture discusses four generations of antivirus technology.

1. Simple Scanner

Uses signatures of known malware.

2. Heuristic Scanner

Looks for suspicious behavior and integrity changes (e.g., checksums or encrypted hashes).

3. Activity Trap

Monitors suspicious program activities.

4. Full-featured Analysis

Modern solutions use:

- Host-based detection.
- Network-based detection.
- Sandboxing.

Comparison of Malware Types

Malware	Needs Host	Self-Replicates	Main Purpose
Trap Door	Yes	No	Unauthorized access
Logic Bomb	Yes	No	Triggered destruction

Trojan Horse	Yes	No	Disguised malicious program
Virus	Yes	Yes	Infect files/programs
Boot Sector Virus	Yes	Yes	Infect boot records
Macro Virus	Yes	Yes	Infect Office documents
Rootkit	Usually	No	Hide malware
Worm	No	Yes	Automatic network spreading
Botnet	No	Controlled by Botmaster	Coordinated attacks
APT	No	Varies	Long-term targeted attack